Tarek Ahmed

SENIOR AEC SOFTWARE DEVELOPER

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SUMMARY

AEC Software Developer sitting at the intersection of engineering and modern software, translating complex real-world challenges into precise, automated digital solutions.

With a background in engineering and a deep passion for software craftsmanship, I've spent years building tools that make BIM workflows faster, smarter, and more reliable, from Revit add-ins and Civil 3D plugins to full-stack platforms connecting design, data, and delivery through intuitive dashboards, deep analysis, and total project visibility.

EDUCATION

B.Sc. Civil Engineering
Ain Shams University
Faculty of Engineering
2013 – 2018
Egypt

LANGUAGES

English	**Fluent**
Arabic	**Native**

Senior AEC Software Developer

Jan 2024 – Present

LTIMindtree Riyadh, Saudi Arabia

Currently deployed as a core member of the Central Development Unit (CDU) - Design Excellence for the Qiddiya giga project, driving digital transformation and enterprise system integration. Past engagements under LTIMindtree include developing Project Management Information System (PMIS) solutions for Qiddiya and delivering customized engineering automation workflows for Langan.

- Leading the architecture and development of enterprise-grade web platforms and automation tools to streamline complex engineering and digital delivery workflows.
- Architecting seamless integrations between centralized project databases and AEC cloud ecosystems (including ACC and APS) to establish a unified, automated data environment.
- Establishing technical standards and optimizing large-scale design review cycles to ensure high-quality project outputs across major client engagements.
- Enhancing cross-disciplinary collaboration and significantly reducing manual engineering overhead by deploying scalable, custom software solutions.

Senior BIM Developer

Jan 2022 – Dec 2023

KEMET Corporation Cairo, Egypt

Creating innovative solutions in the AEC industry to streamline construction workflows and enhance project collaboration.

- Designing, developing, and maintaining software applications that support BIM workflows.
- Collaborating with architects, engineers, project managers, and developers to align software solutions with project objectives and user needs.
- Providing technical guidance and mentorship to team developers, ensuring their expertise in delivering high-quality products.
- Researching and evaluating new technologies to improve BIM workflows.
- Connecting custom AEC software directly with the Autodesk Construction Cloud (ACC) to enable seamless, centralized data exchange.
- Creating custom Revit add-ins using C# and the Revit API to automate repetitive modeling tasks, enforce BIM standards, and accelerate design delivery.
- Developing API integrations with ACC to automate document management, issue tracking, and design review cycles.

BIM Roads Engineer | BIM Developer

Apr 2020 – Dec 2021

Khatib & Alami Cairo, Egypt

As a BIM Roads Engineer, my key responsibilities included implementing the BIM workflow in the design process. Additionally, as a BIM Developer, I was responsible for creating software solutions to enhance the BIM workflow.

- Planning, designing, and grading urban sites.
- Implementing BIM workflow in the design process.
- Creating BIM models for urban sites and checking them for clashes in Navisworks.
- Creating dynamo scripts to automate the design process.
- Creating Civil3D subassemblies using subassembly composer.
- Creating plugins for AutoCAD and Civil3D.

Roads Engineer

Jan 2019 – Mar 2020

Al Amar Consulting Group Cairo, Egypt

As a Roads Engineer, I played a crucial role in the planning and designing of urban sites, ensuring that all projects were executed with a focus on functionality, safety, aesthetics, and adherence to construction standards

- Planning and designing urban sites
- Calculating Bills of Quantities (BOQ)
- Conducting vehicle tracking simulations
- Preparing tender packages

AEC SKILLS

Revit
Navisworks
Civil 3D
AutoCAD
Infraworks
Vehicle Tracking
Dynamo
BIM360
Autodesk Construction Cloud

SOFTWARE DEVELOPMENT SKILLS

C Sharp
DotNet
ASP.net
SQL Server
Entity Framework
Blazor
HTML
CSS
TypeScript
JavaScript
WPF
MVVM Pattern

IMPACT & METRICS

7+
YEARS IN AEC TECH

20+
PLUGINS SHIPPED

10+
ENTERPRISE PROJECTS

7 Apps
PUBLISHED ON AUTODESK APP STORE

Visit TA Solutions on Autodesk App Store

CLICK FOR MORE

FEATURED PROJECTS

CRS Manager

WEB

The CRS Manager is a centralized web application designed to digitize the Comment Resolution Sheet (CRS) process. Operating as a Hybrid Aggregator, it bridges the gap between the technical design data stored in Autodesk Construction Cloud and the contractual negotiation lifecycle. The system implements a Master vs. Mirror pattern where ACC remains the Single Source of Truth for technical data while the CRS app manages the contractual lifecycle, approval statuses, and audit trails. Built on Blazor Server with APS integration, it automatically fetches project issues from ACC, eliminating manual data entry and ensuring data integrity across environments.

Tech Stack

.Net | C# | ASP.Net | Entity Framework | SQL | Blazor

TA Roads Toolkit

PLUGIN

TA Roads Toolkit is a comprehensive Civil3D® solution designed to streamline and automate various tasks in road design, significantly enhancing productivity, reducing errors, and improving the overall quality of road designs. By automating repetitive and time-consuming tasks, TA Roads Toolkit enables engineers to focus on more critical aspects of their projects

Tech Stack

.Net | C# | WPF | AutoCAD APIs | Civil3D APIs | MVVM

MORE PROJECTS

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